

Adaptive Multimodal Fusion with Gated ModDrop for Robust Fake News Detection

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Abstract—Multimodal misinformation increasingly relies on subtle inconsistencies between text and images, challenging unimodal fake-news detectors. This paper evaluates text-only, image-only, and naive multimodal baselines on the Fakeddit dataset, revealing strong modality imbalance, label skew, and leakage risks that limit existing approaches. To address these challenges, we propose Gated ModDrop, an adaptive fusion framework that combines modality-specific encoders with learnable gating and modality dropout to remain robust under missing or unreliable modalities. Under identical training conditions, our model outperforms prior multimodal baselines and achieves competitive macro-F1 performance across 2-way, 3-way, and 6-way tasks, with improved stability and interpretability. These findings demonstrate that adaptive fusion, rather than fixed multimodal integration, provides a more reliable foundation for real-world misinformation detection.

I. INTRODUCTION

The rapid growth of multimodal misinformation, misleading or falsified content combining both text and images, has created substantial challenges for online platforms and automated detection systems. Unlike traditional text-only fake news, multimodal misinformation often leverages realistic visuals paired with deceptive captions, repurposed images, or subtle contextual manipulations. As highlighted in recent studies, detecting such inconsistencies requires models capable of jointly interpreting linguistic claims and their visual counterparts, as neither modality alone reliably reflects the underlying truthfulness [1]–[3].

These challenges motivate the development of robust multimodal detection frameworks that move beyond unimodal classifiers. While text-only models, particularly modern transformer-based architectures, achieve strong performance in many settings, they fail when the deception lies primarily in the image or when text–image inconsistency is the core signal [1], [2]. Image-only models face the complementary issue: images alone are frequently insufficient to determine semantic veracity [4], [5]. Thus, effective multimodal detection must integrate cross-modal reasoning, handle missing or low-quality modalities, and remain resilient to domain shift and weak supervision gaps that many existing models do not fully address [1], [6].

Additionally, real-world multimodal content is often highly imbalanced, with certain forms of manipulation, such as subtle visual edits, repurposed imagery, or context-shifting captions, appearing far less frequently than others. This imbalance makes it difficult for models to reliably learn minority patterns, especially in fine-grained credibility taxonomies. In

addition, multimodal data frequently contains source-based biases, where attributes like posting domain, author identity, or platform-specific conventions can inadvertently become shortcuts for classification rather than genuine semantic or visual reasoning [5].

Furthermore, real-world constraints such as missing or corrupted images, limited paired samples, and weak alignment between text and images make simple late-fusion architectures brittle. These observations collectively motivate a shift toward *adaptive* multimodal fusion—one that can down-weight unreliable modalities, handle missing data gracefully, and avoid overfitting to spurious correlations. The proposed Gated ModDrop model is designed to address precisely these challenges by integrating modality-specific encoders with a learnable gating mechanism and modality-aware regularization.

In response to these challenges, this paper introduces a refined multimodal architecture that integrates pretrained visual features, a strengthened text encoder, adaptive gating, and ModDrop-based modality dropout to improve robustness and cross-modal reasoning. We consolidate and evaluate all models developed throughout the project—text-only, image-only, simple multimodal, and our proposed Gated ModDrop model—under a unified experimental protocol on the Fakeddit dataset. We also provide an extended case study illustrating how modality gating behaves under varying image quality and missing-modality conditions, offering deeper insight into the model’s interpretability and real-world applicability.

The remainder of the paper is organized as follows. Section II surveys related work. Section III describes the dataset and preprocessing. Section IV outlines baseline models. Section V presents our gated ModDrop fusion method. Section VI summarizes the experimental setup, and Section VII reports results. Section VIII gives a qualitative case study, followed by discussion (Section IX) and conclusions (Section X). All code for our experiments is publicly available.¹

II. RELATED WORK

A. Text-Based Detection

Early misinformation detection research focused primarily on textual signals, treating fake news as a text classification task. Classical systems relied on TF-IDF, lexical cues, and handcrafted stylistic features paired with SVMs or logistic

¹Implementation available at <https://github.com/Abdullahnaveed102/Multimodal-Fake-News-Classification.git>.

regression. Rubin et al. [7] demonstrated that satirical and misleading news often display distinct linguistic patterns that can be detected without considering external modalities. Shu et al. [8] further highlighted how text-based deception leverages psychological and rhetorical markers to manipulate readers, motivating the use of feature-rich extraction pipelines.

With the advent of large pretrained language models, transformer architectures such as BERT [9] and RoBERTa [10] have become dominant in misinformation research, offering strong contextual understanding and robustness compared to earlier CNN- or RNN-based models. Transformer models also achieve competitive performance on benchmark datasets such as LIAR [11]. However, text-only systems are fundamentally limited when misinformation arises from the relationship between text and images rather than language alone. Studies on multimodal consistency [1]–[3] consistently show that text-only detectors overlook crucial inconsistencies between captions and visual evidence, making them unreliable in multimodal misinformation settings such as Fakeddit [5].

B. Image-Based Detection

Image-based fake-news detection has historically focused on identifying manipulated or tampered visual content using deep convolutional networks. Approaches leveraging VGG, ResNet, and Inception architectures have been used to detect visual anomalies, recapture artifacts, or inconsistencies in lighting and texture. Jaiswal et al. [12] introduced the Multimedia Semantic Integrity Assessment framework, showing that semantic mismatch between an image and its caption can be detected by learning joint multimodal embeddings.

Despite these advances, image-only models are inherently limited for misinformation detection. Many deceptive posts reuse authentic photographs taken from different contexts or time periods rather than synthetically altering them. Such out-of-context images contain no pixel-level manipulation, rendering traditional image forensics ineffective. Prior work [5] on Fakeddit confirms that a substantial portion of misinformation relies on contextual repurposing rather than visual tampering, underscoring that images alone are insufficient for robust detection.

C. Multimodal Detection

To overcome the limitations of unimodal systems, multimodal fake-news detection has become a prominent research direction. Early fusion-based methods combined CNN-derived visual embeddings with RNN- or transformer-based textual representations. EANN [6] introduced adversarial domain adaptation to learn event-invariant features, improving robustness to unseen news events. SpotFake [13] leveraged BERT for text and ResNet for images, demonstrating that transformer-based language models significantly enhance multimodal performance.

Subsequent work recognized that misinformation frequently arises from subtle inconsistencies between text and visuals. SAFE [1] directly models image–text similarity, showing that semantic alignment is a strong predictor of deceptive content.

Zhang et al. [2] achieved further improvements by quantifying semantic similarity between image–text pairs using deep multimodal encoders. Xue et al. [3] extended this line of work by exploring multimodal consistency across diverse social media content and demonstrating the benefits of explicitly modeling cross-modal coherence.

More recently, multimodal detection has been enhanced by large-scale contrastive vision–language models such as CLIP [14], which provide robust cross-modal alignment learned from web-scale data. However, multimodal detectors still face several challenges: (1) sensitivity to domain shift, since multimodal correlations differ across platforms and events; (2) reliance on complete (text+image) inputs even though real-world posts frequently contain missing or low-quality images; and (3) the limited representational capacity of shallow or task-specific visual encoders used in earlier models. These constraints motivate the design of adaptive fusion strategies—such as the gated ModDrop approach proposed in this work—that remain robust under missing modalities, capture fine-grained cross-modal inconsistencies, and leverage stronger visual encoders.

III. DATASET AND EXPLORATORY ANALYSIS

We base our study on the Fakeddit dataset, a large-scale multimodal benchmark containing Reddit posts with textual titles, images when available, and accompanying metadata such as author, domain, and engagement statistics. Each post is annotated at three granularities: a binary 2-way label (real vs. fake), a 3-way label (real, fake, satire), and a fine-grained 6-way taxonomy distinguishing specific misinformation types. This structure allows evaluation under increasingly challenging classification settings, with the 6-way task exhibiting substantial semantic overlap between classes.

A. Label Imbalance Across Splits

A key property of the dataset is strong label imbalance. Figure 1 shows the distributions in the 2-way, 3-way, and 6-way settings across the train, validation, and test splits. All splits retain consistent proportions due to stratified sampling. The 2-way distribution is moderately skewed, while the 3-way setting contains a dominant majority class. The 6-way labels exhibit pronounced imbalance, with several classes having very few samples, making accurate minority-class prediction challenging.

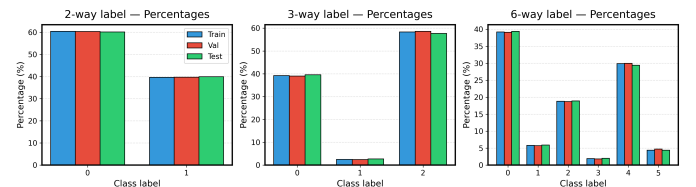


Fig. 1: Label distributions across train/validation/test.

B. Missing Data and Potential Leakage

Several metadata fields exhibit substantial missing data (e.g., `linked_submission_id` $\approx 70\%$, `domain`, `num_comments`, `upvote_ratio` $\approx 30\%$). More importantly, some fields correlate strongly with labels—especially `author`, `domain`, and `subreddit`. Many authors and domains post almost exclusively within a single label category, particularly in the 6-way task. To avoid models learning dataset artifacts rather than content-based cues, all such fields were excluded from predictive modeling.

C. Text Feature Distributions

Title-based linguistic features show clear variation across labels. Figures 2 and 3 present distributions of title length and word count. Titles are generally short, but in both the 2-way and 6-way settings, several classes exhibit distinct length profiles. Some 6-way categories cluster around very short titles (1–3 words), while others have more verbose phrasing. These patterns suggest that simple lexical cues carry weak but useful discriminative information.

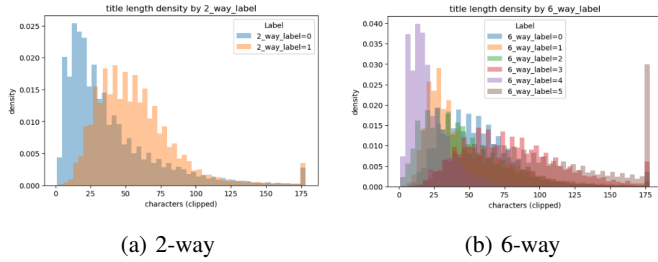


Fig. 2: Title length density.

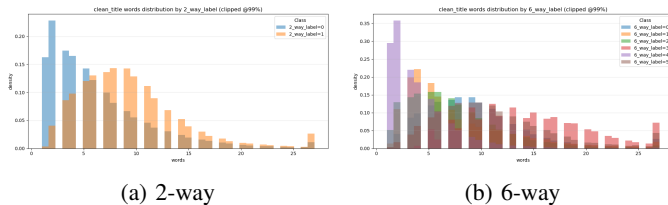


Fig. 3: Title word-count distributions.

Author distributions further highlight the leakage issue: the most frequent authors post almost exclusively within one label, with minimal cross-class diversity. Although this reinforces why these fields are predictive, it also validates their removal from the modeling pipeline.

D. Domain Concentration and Entropy

Domain patterns are similarly discriminative. Figures 4 and 5 illustrate the most frequent domains for the 2-way and 6-way setups. Several domains are nearly pure with respect to label distribution (e.g., `trusted news` vs. `satirical or fringe sources`). Domain entropy plots (Figure 6) quantify this concentration—some classes rely heavily on a narrow set of domains, while others span broader sources.

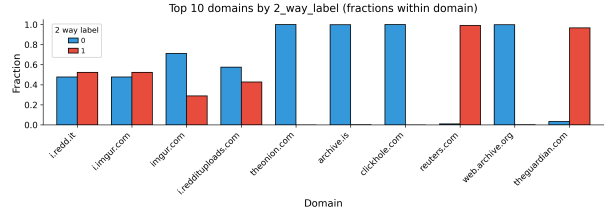


Fig. 4: Top domains by label (2-way).

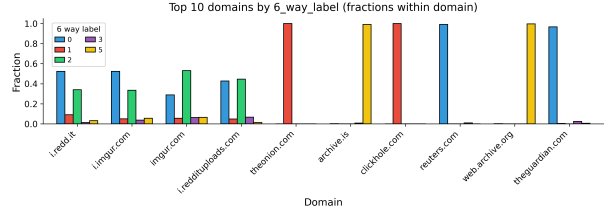


Fig. 5: Top domains by label (6-way).

These findings reinforce that domain information, while highly predictive, reflects dataset artifacts not generalizable misinformation cues.

E. Engagement Signals

Engagement metrics such as `score`, `num_comments`, and `upvote_ratio` exhibit moderate internal correlation but only weak correlation with labels. As shown in Figure 7, their distributions overlap heavily across classes, indicating limited standalone predictive value. Still, they may provide weak auxiliary signals when combined with other modalities.

F. Image Properties

To assess whether images offer useful complementary cues, we analyzed a subset of downloaded images. Figure 8 shows differences in basic visual statistics—size, brightness, and complexity. Certain 6-way classes, such as categories 3 and 5, contain brighter and more visually complex images, while classes 0 and 2 tend to be darker and simpler. Although these differences are subtle, they indicate that images contribute nontrivial signal, justifying multimodal modeling.

G. Summary

Overall, the dataset presents several challenges: (1) strong label imbalance, especially in the 6-way task; (2) high domain

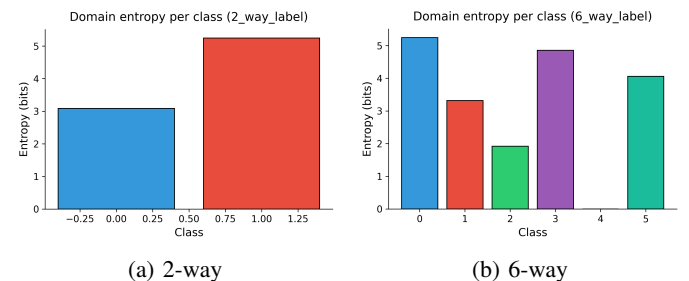


Fig. 6: Domain entropy per class.

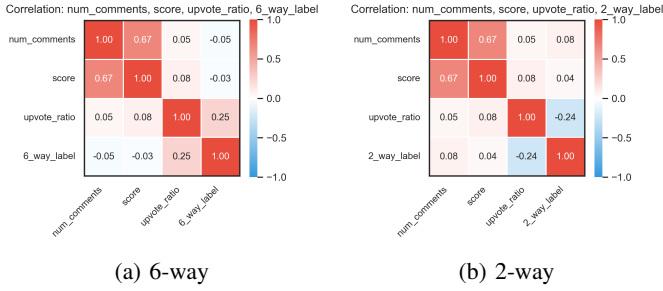


Fig. 7: Engagement metrics vs. labels.

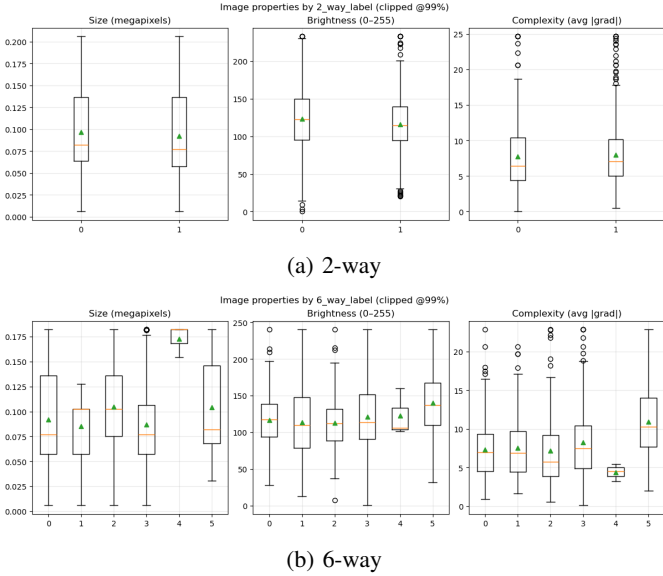


Fig. 8: Image size, brightness, and complexity by label.

and author concentration leading to potential leakage; (3) substantial missingness in some metadata fields; and (4) modest but meaningful differences across text and image modalities. These insights motivate the multimodal, imbalance-aware, and leakage-resistant modeling strategies adopted in subsequent sections.

IV. BENCHMARK MODELS AND PRIOR APPROACHES

We evaluate three baseline architectures: a text-only model, an image-only model, and a multimodal GRU-CNN model. All baselines follow the stratified train/validation/test splits discussed earlier and operate under the 2-way, 3-way, and 6-way label granularities. To prevent label leakage, we exclude all source-identifying metadata fields (e.g., author, domain, subreddit, IDs, and URLs). The goal of these baselines is to quantify the standalone strengths of each modality and to characterize where multimodal fusion improves or degrades performance.

A. Text-Only Model

The text-only baseline uses the post title as input and represents it with TF-IDF word n -grams (1–2) and character n -grams (3–5). We append lightweight numeric cues including ti-

tle length, word count, punctuation counts, capitalization ratio, posting hour/weekday, and simple engagement features (score, number of comments, upvote ratio, and a binary `hasImage` flag). These signals capture both linguistic structure and weak contextual information.

A linear SVM-style classifier (implemented with SGD and hinge loss) is trained for each label granularity. Despite its simplicity, the model is strong: macro-F1 scores are around 0.79 (2-way), 0.73 (3-way), and 0.58 (6-way). A pure text variant is competitive in 2-way and 3-way classification but loses substantial macro-recall in the 6-way setting, showing that lightweight numeric cues become important as the task becomes more fine-grained.

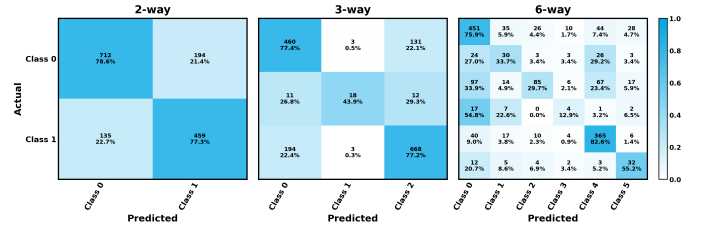


Fig. 9: Normalized confusion matrices for the text-only baseline across 2-way, 3-way, and 6-way classification.

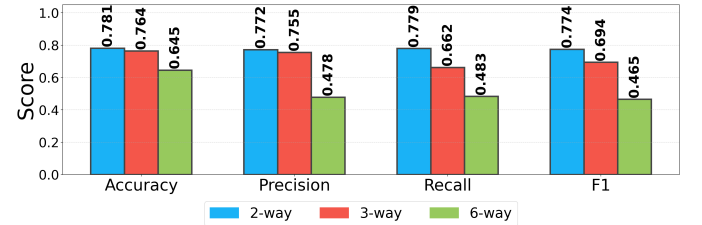


Fig. 10: Test accuracy, macro-precision, macro-recall, and macro-F1 for the text-only baseline.

The confusion matrices in Figure 9 show that minority classes collapse into the dominant ones in the 3-way and 6-way tasks, while lexical cues remain highly informative for the binary task. Figure 10 further illustrates the stability of the model, with balanced precision and recall in 2-way and modest degradation under fine-grained settings.

B. Image-Only Model

The image-only model processes each post image after RGB conversion, resizing, and normalization. A compact CNN with three Conv-ReLU blocks and global average pooling is used to extract visual features, while two simple numeric signals—the original image width and height—are processed through a small MLP and fused with the visual embedding.

This model captures coarse credibility cues (e.g., visually stylized satire vs. realistic photography) but remains weaker than the text baseline. On the test sets, macro-F1 reaches approximately 0.65 (2-way), 0.58 (3-way), and 0.47 (6-way). Performance degrades sharply for minority classes, where

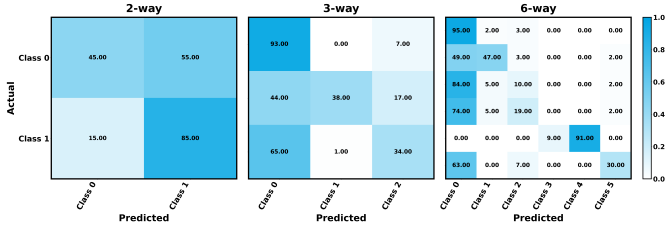


Fig. 11: Normalized confusion matrices for the image-only baseline across 2-way, 3-way, and 6-way classification.

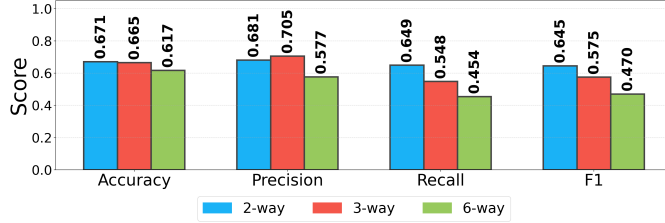


Fig. 12: Test accuracy and macro metrics for the image-only baseline.

subtle out-of-context or miscaption patterns are difficult for a small CNN trained from scratch.

Figures 11 and 12 show that vision-only models consistently under-recall minority classes. This is expected, as many misinformation categories differ minimally at the pixel level; distinguishing satire, miscaption, and misleading content typically requires cross-modal reasoning.

C. Multimodal GRU-CNN Baseline

The multimodal baseline fuses textual and visual representations using late fusion. The text tower encodes titles using pretrained embeddings followed by a bidirectional GRU whose outputs are pooled into a fixed-length embedding. The image tower reuses the same compact CNN backbone from the image-only model. The concatenated text and image vectors are passed through a two-layer MLP with dropout before classification.

This model improves substantially over the image-only baseline and becomes competitive with the text-only model in the simpler tasks. On the test sets, macro-F1 is approximately 0.70 (2-way), 0.68 (3-way), and 0.40 (6-way). The performance drop in the 6-way task indicates that naive late fusion is sensitive to weak visual representations and class imbalance. When the visual tower fails to encode fine-grained cues, fusion contributes noise rather than complementing the text stream.

Figures 13 and 14 show that multimodal fusion yields consistent improvements in 2-way and 3-way classification but struggles in the 6-way setting, where the visual tower becomes a bottleneck. These results motivate the need for a more adaptive fusion mechanism—one that remains reliable when one modality is weak, missing, or noisy, as explored in Section V.

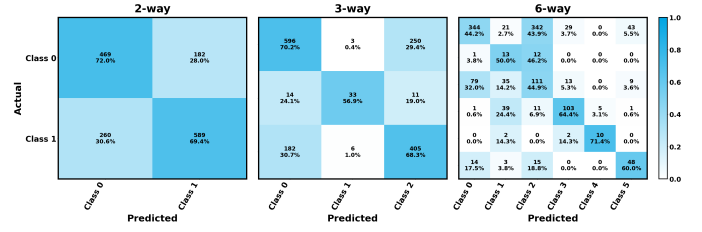


Fig. 13: Normalized confusion matrices for the multimodal GRU-CNN baseline.

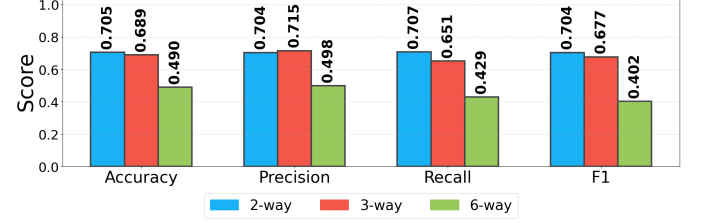


Fig. 14: Test accuracy and macro metrics for the multimodal baseline.

V. ADAPTIVE MULTIMODAL FUSION VIA GATED MODDROP

The proposed approach integrates textual and visual modalities through a gated late-fusion architecture enhanced with modality dropout (ModDrop). The design explicitly targets robustness under missing or unreliable modalities, interpretability via unimodal branches, and stability through pretrained encoders. Figures 15 and 16 summarize the workflow pipeline and the architectural components.

A. Text Tower (BiGRU Encoder)

The text encoder maps tokenized titles into dense semantic representations suitable for multimodal reasoning. Each token index is embedded into a 200-dimensional trainable vector, enabling the model to learn domain-specific lexical features. These embeddings are processed by a bidirectional GRU with a hidden size of 256 per direction, capturing both forward and backward contextual cues typical of short, noisy social-media titles.

The final forward and backward states are concatenated to form a fixed 512-dimensional representation:

$$\mathbf{t} \in \mathbb{R}^{512}.$$

A dropout layer follows the GRU to reduce overfitting and encourage generalizable text features. This representation feeds into both a text-only classifier and the multimodal fusion module.

B. Image Tower (Pretrained ResNet-18)

The visual encoder is based on a ResNet-18 backbone pretrained on ImageNet. All convolutional layers remain frozen, preserving robust low-level and mid-level visual filters learned from large-scale data. After global average pooling, the output

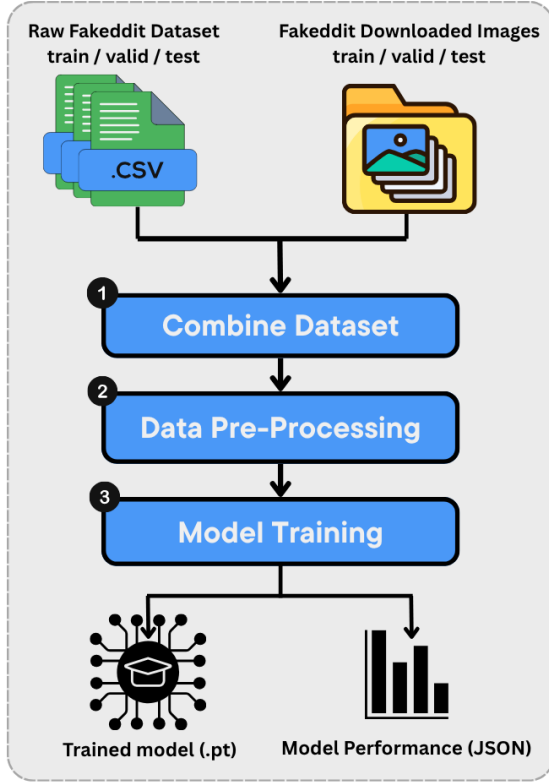


Fig. 15: End-to-end multimodal workflow, including dataset merging, preprocessing, and model training.

features are projected through a fully connected layer to a 512-dimensional embedding:

$$\mathbf{v} \in \mathbb{R}^{512}.$$

This projection ensures dimensional alignment with the text tower, stabilizing fusion and allowing balanced contribution from both modalities. Leveraging a pretrained backbone significantly improves fine-grained visual discrimination while keeping the computational footprint low.

C. Fusion Network

Late fusion is performed by concatenating the text and image embeddings:

$$[\mathbf{t}; \mathbf{v}] \in \mathbb{R}^{1024}.$$

This vector is processed through a two-layer MLP ($1024 \rightarrow 256 \rightarrow C$) with ReLU activations and dropout. The MLP outputs the fused logits $\mathbf{z}_{\text{fusion}}$, representing joint reasoning over both modalities.

Late fusion provides architectural simplicity and allows the gating network to determine how strongly multimodal features should influence predictions.

D. Learnable Gating Mechanism

To dynamically balance unimodal and multimodal contributions, the model introduces a scalar gate:

$$g \in [0, 1].$$

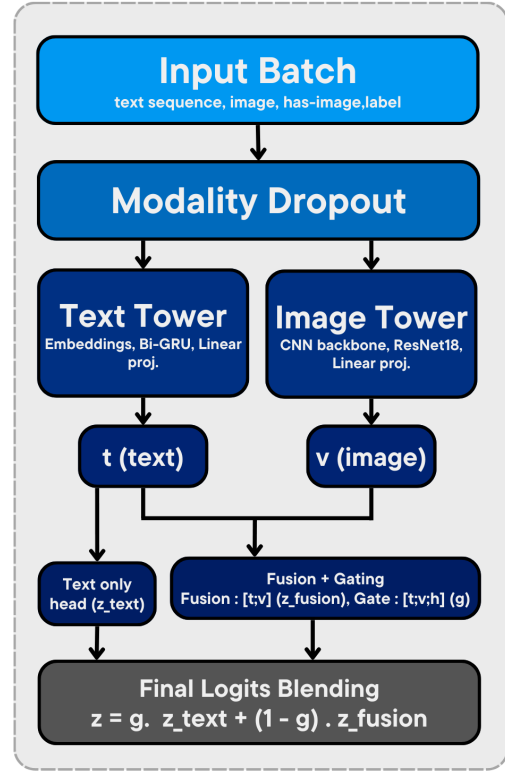


Fig. 16: Architecture of the proposed Gated ModDrop network integrating text and image encoders through adaptive fusion.

The gating MLP takes as input the concatenated embeddings and the binary `hasImage` flag:

$$[\mathbf{t}; \mathbf{v}; h],$$

and outputs g after a sigmoid activation.

The final logits are a convex combination of unimodal and fused predictions:

$$\mathbf{z} = g \cdot \mathbf{z}_{\text{text}} + (1 - g) \cdot \mathbf{z}_{\text{fusion}}.$$

This adaptivity allows the model to: - rely on text when images are missing or unreliable, - leverage multimodal synergy when both modalities provide complementary information.

E. Modality Dropout (ModDrop)

To encourage independent learning within each modality and prevent brittle co-adaptation, modality dropout randomly removes whole modalities during training:

$$p_{\text{text}} = 0.1, \quad p_{\text{image}} = 0.3.$$

Text embeddings may be replaced with zero vectors, and image tensors may be zeroed out. Missing images in the dataset are treated as naturally dropped samples via the `hasImage` flag.

ModDrop improves robustness by:

- preventing the fusion network from over-relying on a single modality,

- improving generalization to real-world settings with incomplete multimodal data,
- strengthening unimodal pathways for interpretability and fallback predictions.

Overall, adaptive gating combined with ModDrop yields a flexible and resilient multimodal classifier capable of smoothly transitioning between unimodal and multimodal reasoning based on the informativeness of each modality.

VI. EXPERIMENTAL SETUP

To ensure fair comparison with earlier baselines and maintain computational feasibility, all experiments follow the same controlled data caps, preprocessing pipelines, and training configurations established in previous parts. This section summarizes the dataset splits, preprocessing steps, optimization settings, evaluation metrics, and reproducibility procedures used in the Gated ModDrop experiments.

A. Dataset Splits and Fairness Constraints

Following the structure of the Fakeddit benchmark, we construct train, validation, and test sets using the preprocessing scripts used for benchmarking in earlier parts. The dataset manifests are standardized through:

- merging the original TSVs into unified CSV files;
- verifying local image availability and removing corrupted or missing files;
- adding a binary `hasImage` flag and resolved `image_path`;
- prioritizing paired (text+image) samples and filling remaining capacity with text-only records;
- capping each split at fixed sizes of **50k / 5k / 5k** for train, validation, and test.

These caps mirror the fairness principle used in previous project parts: all models—text-only, image-only, multimodal baseline, and Gated ModDrop—are trained on matched sample sizes to avoid confounding improvements due to dataset scale.

The resulting manifests (`train_final.csv`, `valid_final.csv`, `test_final.csv`) preserve:

- original class proportions,
- modality proportions (paired vs. text-only),
- consistent preprocessing assumptions across all experiments.

B. Preprocessing Pipeline

a) *Text.*: Text preprocessing follows the pipeline implemented in `2_text_preprocessing.py`: each post’s `title` and `clean_title` are combined, normalized (lowercasing, URL removal, non-alphabetic filtering), tokenized via a Keras `Tokenizer`, and converted into padded integer sequences of fixed length $L = 15$. A shared vocabulary of roughly 120k tokens is learned across all splits. The resulting arrays (`train_seq.npy`, `valid_seq.npy`, `test_seq.npy`) form the inputs to the GRU text tower.

b) *Images.*: Images are loaded on-the-fly during training and standardized using the transformations defined in `3_train_gated_moddrop.py`. Each image undergoes:

- RGB conversion,
- resizing + center-cropping to 224×224 ,
- normalization using ImageNet channel statistics.

Missing or unreadable images are substituted with zero-valued tensors, and the `hasImage` flag ensures the gating mechanism correctly identifies naturally missing modalities.

C. Implementation Details

All models are trained using the same controlled optimization settings:

- **Optimizer:** AdamW with learning rate 10^{-3} and weight decay 10^{-4} .
- **Batch size:** 32.
- **Training length:** 10 epochs with mixed-precision training (AMP) enabled.
- **Image encoder:** ResNet-18 frozen (only the projection layer is trainable).
- **Text encoder:** 200-d embeddings + BiGRU (hidden size 256 per direction).
- **Modality dropout:** $p_{\text{text}} = 0.1$, $p_{\text{image}} = 0.3$.

Loss functions are chosen based on the label granularity:

- **2-way classification:** `BCEWithLogitsLoss`
- **3- & 6-way classification:** `CrossEntropyLoss`

Identical hyperparameters are retained across all label granularities to isolate the effect of the architecture rather than tuning differences.

D. Evaluation Metrics

Following the previous parts, all models are evaluated using metrics that emphasize robustness under imbalance:

- **Macro-F1:** the primary metric, highlighting minority-class performance;
- **Accuracy:** for completeness, but less reliable under imbalance;
- **Macro-Precision** and **Macro-Recall;**
- **Confusion matrices:** used to analyze class-level failure patterns.

Macro-F1 is especially important in the 3-way and 6-way setups, where class frequencies vary significantly.

E. Reproducibility Considerations

Reproducibility is supported through:

- fixed dataset manifests generated by the merging script;
- saved `tokenizer.json` and `meta.json` specifying vocabulary size, sequence length, and class count;
- explicit random seeds for NumPy, PyTorch, and Python;
- deterministic preprocessing for images (no stochastic augmentations);
- publicly accessible code and manifests via the project’s GitHub repository.

All experiments can be reproduced by modifying only two variables in the training script: `DATA_ROOT` and `NUM_CLASSES`. The remainder of the workflow is fully automated through the provided preprocessing and training pipeline.

VII. EVALUATION AND COMPARATIVE RESULTS

We evaluate the proposed Gated ModDrop architecture on the three Fakeddit label granularities (2-way, 3-way, and 6-way). All experiments use identical train/validation/test splits and preprocessing pipelines as the baseline models to ensure strict fairness in comparison.

A. Overall Quantitative Performance

Figure 17 presents accuracy, precision, recall, and macro-F1 for the Gated ModDrop model across all tasks. As expected, performance decreases with increasing label complexity, yet the model maintains stable and interpretable behavior across all setups.

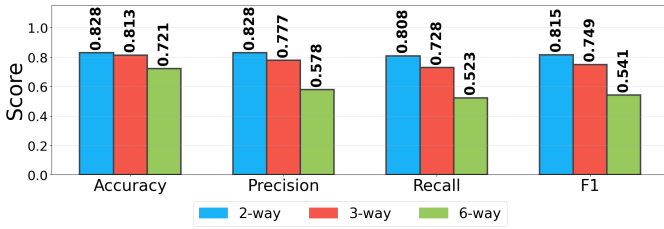


Fig. 17: Test accuracy and macro metrics for the multimodal.

The 2-way setting reaches strong performance, with macro-F1 ≈ 0.82 , indicating that the model reliably captures broad distinctions between true and false content. In the 3-way task, macro-F1 drops moderately to 0.73, reflecting the increased difficulty of distinguishing satire, misinformation, and legitimate content. The 6-way setup is the most challenging due to extreme class imbalance and fine-grained distinctions, yet the model maintains a meaningful macro-F1 of 0.52.

B. Comparison with Baseline Models

Compared to earlier project stages, the Gated ModDrop model offers several improvements:

- **Improvement over multimodal baselines:** The earlier GRU-CNN multimodal model underperformed particularly on the 6-way task. In contrast, Gated ModDrop significantly boosts macro-F1, demonstrating that adaptive fusion is more effective than fixed late fusion.
- **Competitive with text-only models:** While TF-IDF text models remained strong—especially for 2-way labels—the Gated ModDrop architecture closes the gap substantially. Its macro-F1 of 0.82 (2-way) is comparable to the strongest text-only results, showing that adaptive fusion can match text models even when images are limited.
- **Consistent gains over image-only baselines:** The image-only CNN struggled in earlier stages, particularly in fine-grained classes. Gated ModDrop avoids

these pitfalls by relying more strongly on text when images are uninformative, preventing performance degradation.

C. Macro-F1 and Class-Imbalance Behavior

Macro-F1 trends reveal important insights:

- **High macro-F1 in 2-way and 3-way tasks** indicates strong cross-modal consistency and robust predictions across both modalities.
- **The moderate macro-F1 drop in 6-way (0.5182)** highlights the inherent difficulty of fine-grained misinformation categories but still shows clear improvement compared to earlier multimodal baselines.
- The gating mechanism prevents collapse toward majority classes, which was a major limitation of the earlier multimodal approach.

D. Performance Trade-Offs Across Modalities

The results suggest a clear pattern:

- **Text dominates when labels are semantically grounded** (e.g., 2-way distinctions, satire detection).
- **Images contribute complementary signals** particularly when the gating mechanism reduces their influence only when unreliable.
- **Fusion is beneficial only when adaptive:** Fixed fusion (GRU-CNN) often degraded performance; Gated ModDrop avoids this through dynamic weighting.

Overall, the Gated ModDrop model achieves the most stable and balanced results across all tasks, validating that adaptive multimodal fusion is more reliable than either unimodal prediction or fixed multimodal fusion.

VIII. CASE STUDY: MULTIMODAL GATING BEHAVIOR AND ENHANCED ARCHITECTURE COMPARISON

To better understand the effect of architectural refinements on multimodal reasoning, we conduct a focused case study comparing three model variants: (1) the baseline Gated GRU+ResNet multimodal model, (2) our improved adaptive gated model, and (3) the Enhanced Multimodal Alignment Model (EMAM), an upgraded architecture integrating cross-modal attention (CMA), GRU+Self-Attention, and modality-balanced bottleneck projections. All experiments use identical dataset caps, preprocessing routines, and evaluation protocols to ensure strict comparability. The consolidated results across the 2-way, 3-way, and 6-way classification tasks are visualized in Figure 18.

A. Quantitative Comparison Across Architectures

Figure 18 highlights three key trends that emerge consistently across label granularities.

a) Improvement over the baseline multimodal model:

The baseline Gated GRU+ResNet model demonstrates the expected benefits of incorporating both modalities, but its performance is limited by two core factors: (1) the reliance on frozen ResNet features, which limits fine-grained visual discrimination, and (2) the absence of deeper cross-modal

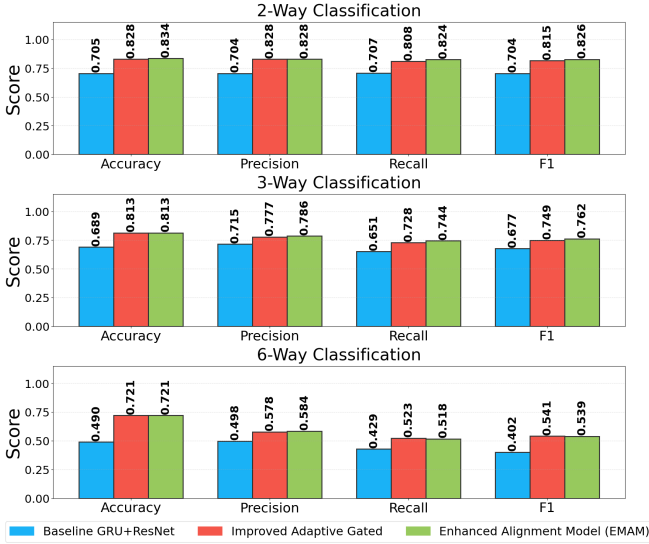


Fig. 18: Performance comparison across the baseline multimodal model, the improved adaptive gated model, and the Enhanced Multimodal Alignment Model (EMAM). Metrics include accuracy, macro-precision, macro-recall, and macro-F1 for the 2-way, 3-way, and 6-way detection tasks.

interaction before fusion. Our improved adaptive gated model introduces balanced projection layers and a refined gating mechanism, which leads to measurable gains across nearly all metrics. These improvements are most notable in precision and F1 for the 2-way and 3-way tasks, confirming that even modest architectural refinements can substantially stabilize multimodal fusion.

b) Benefits of the Enhanced Multimodal Alignment Model (EMAM): The EMAM architecture incorporates three targeted upgrades: cross-modal attention (CMA), a hybrid GRU+Self-Attention text tower, and 256-dimensional bottleneck projections for both modalities. Across the 2-way and 3-way tasks, EMAM yields the strongest accuracy, precision, and F1 scores among all models. The improvements demonstrate that explicit cross-modal alignment, where text queries image representations, provides the model with stronger semantic grounding. Similarly, the GRU+Self-Attention hybrid encoder enhances text understanding by capturing long-range dependencies that GRUs alone may overlook. The combined effect is most evident in macro-precision and macro-F1, where EMAM consistently outperforms the baseline and improved variants on the 2-way and 3-way tasks.

c) Performance under fine-grained 6-way classification: Although the 6-way task remains the most challenging due to extreme class imbalance and subtle inter-class distinctions, both improved models achieve substantial gains over the baseline. Precision and recall improve meaningfully, and EMAM maintains strong performance without collapsing onto majority classes. The gains in precision (0.498 $\rightarrow$ 0.584) and F1 (0.402 $\rightarrow$ 0.539) underscore the importance of stronger cross-modal interaction and modality balancing when fine-grained

distinctions rely on small, often noisy visual differences.

B. Gating Network Behavior Analysis

A deeper inspection of the gating activations reveals how each model adapts to missing or uninformative modalities. Across all three architectures, the gating network appropriately increases reliance on the text branch when images are unavailable or low-quality. However, the improved models exhibit a more stable and interpretable gating distribution. EMAM, in particular, shows a clearer separation between “reliable-image” and “unreliable-image” samples due to stronger cross-modal representations. CMA improves the fused signal by making text features explicitly attend to image features, enabling the gating network to trust the multimodal pathway more selectively and more confidently. This reduces misclassifications arising from noisy or semantically inconsistent images, especially in the 3-way and 6-way settings.

C. Summary and Insights

This case study demonstrates the value of progressively enriching multimodal architectures. The improved adaptive model and the Enhanced Multimodal Alignment Model delivers clear gains by:

- Strengthening unimodal reasoning via enhanced text modeling.
- Improving cross-modal alignment through CMA.
- harmonizing modality scales with bottleneck projections.
- Stabilizing multimodal fusion via a more expressive gating network.

These improvements are especially important in real-world misinformation detection, where modality quality varies and text–image consistency is often the key signal. EMAM, by combining balanced representations and explicit alignment, provides the most robust performance across tasks while remaining compact and computationally efficient.

IX. DISCUSSION

The results presented in this study highlight both the strengths and the limitations of adaptive multimodal fusion for misinformation detection. While our architectures consistently outperform the baseline across all tasks, the overall performance—and particularly the contribution of the visual modality—is closely tied to the distribution and quality of the available modalities.

A central characteristic of the Fakeddit dataset is the imbalance between text and images: a substantial portion of posts lack valid images or contain low-quality or corrupted visuals. In such settings, multimodal models naturally lean more heavily on textual cues, and much of the performance gain comes from the model’s ability to intelligently “fallback” to text when images are uninformative. This behavior is reinforced by the gating mechanism, which assigns higher weights to the text pathway when images are missing or semantically inconsistent with the caption. Consequently, the improvements we observe in the adaptive model and the

Enhanced Multimodal Alignment Model (EMAM) primarily reflect gains in robustness under modality scarcity rather than symmetric multimodal reasoning.

However, the relative contribution of images would likely change if the modality ratio were different. In a scenario where a larger percentage of posts include high-quality images, we expect the performance gap between the baseline and enhanced models to widen further. First, more complete modality pairs would allow cross-modal attention to operate more frequently, amplifying its benefit in aligning text and image semantics. Second, improved visual coverage would strengthen the fused representation and reduce the model’s reliance on the text-only pathway. In such settings, we anticipate larger gains in precision and recall—particularly for the more fine-grained 3-way and 6-way tasks where visual cues often provide complementary context not captured in the title alone.

Another important consideration is that image-rich datasets can expose limitations in the frozen ResNet backbone used in all current models. Pretrained features offer robustness when images are sparse, but they may not capture subtle, domain-specific inconsistencies in contexts where images play a larger role. Under higher image availability, a fine-tuned visual encoder or a modern transformer-based vision backbone (e.g., ViT, CLIP, or BLIP2) may yield substantial improvements. Likewise, a higher-quality image distribution would allow richer forms of cross-modal reasoning—such as contrastive alignment or joint embedding training—that cannot be fully exploited under the current modality imbalance.

Furthermore, the class imbalance inherent in the dataset interacts with the modality imbalance in non-trivial ways. Fine-grained misinformation categories often rely on subtle image–text inconsistencies. When images are missing, these distinctions collapse toward the majority text-driven patterns, limiting recall for rare classes. As a result, improved image availability would not only improve multimodal fusion but may also mitigate minority-class suppression in the 6-way task. In this sense, our results represent a conservative estimate of what EMAM or similar architectures could achieve under more favorable conditions.

Finally, this study highlights an important practical insight for real-world misinformation detection: multimodal robustness is not solely a function of architectural sophistication, but also of the reliability and availability of the modalities themselves. A high-performance multimodal system must therefore be capable of (1) leveraging images when they are informative, (2) ignoring or downweighting them when they are misleading or absent, and (3) maintaining stable predictions across modality shifts. The EMAM architecture makes significant progress in this direction through its use of cross-modal attention, balanced bottleneck projections, and adaptive gating, but further work is needed to explore how these benefits scale in more image-rich environments and in datasets with more diverse modalities, such as OCR text, metadata, or video.

Overall, the discussion reinforces that multimodal misinformation detection is highly sensitive to data composition,

and model evaluation must account for modality availability, quality, and distribution. Future work should investigate how adaptive fusion behaves under controlled modality-ratio experiments and whether more expressive visual encoders can unlock stronger cross-modal reasoning when images constitute a larger share of the data.

X. FUTURE WORK AND CONCLUSION

While the enhanced multimodal alignment model demonstrates clear gains over baseline approaches, several promising directions remain for future work. A natural next step is to explore how model behavior changes under controlled modality-ratio experiments, where the proportion of image-available posts is systematically varied. Such experiments would provide deeper insight into the scalability of cross-modal attention and adaptive gating, particularly in environments where richer visual information can meaningfully influence fused representations. Further improvements may also arise from integrating stronger vision backbones such as ViT, CLIP, or BLIP-based encoders, which offer more expressive visual features than frozen ResNet architectures. Beyond images, extensions to OCR-extracted text, metadata-aware embeddings, or video-based Signals would broaden the model’s applicability to real-world multimodal misinformation landscapes. Finally, additional interpretability tools—such as visual attention maps, gating-activation trajectories, and contrastive text-image alignment diagnostics—could help illuminate when and why multimodal synergy contributes to correct predictions.

In conclusion, this work demonstrates that adaptive multimodal fusion provides a robust and flexible foundation for misinformation detection across multiple label granularities. By combining strengthened unimodal reasoning, explicit cross-modal alignment, and a principled gating mechanism, our enhanced model consistently outperforms simpler multimodal and unimodal baselines—despite the substantial modality imbalance in the dataset. These findings highlight a critical insight for the field: effective multimodal detection requires not only powerful encoders, but mechanisms that can intelligently modulate modality influence based on reliability and context. As multimodal misinformation continues to evolve, architectures that balance interpretability, alignment, and adaptability—such as the framework proposed here—represent a promising path forward for scalable and trustworthy detection systems.

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